

Cheat Sheet for MDP task

1. Original Bellman equation for MDP

$$U(s) = \max_a \sum_{s'} P(s'|s, a) [R(s, a, s') + \gamma U(s')] \quad \left. \vphantom{\sum_{s'}} \right\} \text{non-linear system}$$

$$\pi^*(s) = \operatorname{argmax}_a \sum_{s'} P(s'|s, a) [R(s, a, s') + \gamma U(s')]$$

$$\text{let } Q(s, a) = \sum_{s'} P(s'|s, a) [R(s, a, s') + \gamma U(s')]$$

$$\text{then } U(s) = \max_a Q(s, a) \quad \pi^*(s) = \operatorname{argmax}_a Q(s, a)$$

2. Bellman update used by value iteration

$$U_{i+1}(s) = \max_a \sum_{s'} P(s'|s, a) [R(s, a, s') + \gamma U_i(s')]$$

- if you define a general Q function in your program as $Q(\text{mdp}, s, a, U)$

$$\text{then } U_{i+1}(s) = \max_a Q(\text{mdp}, s, a, U_i)$$

- set $U_0(s) = 0$ for all states!

3. One-step look-ahead policy $\rightarrow$ also used by your policy iteration's policy improvement step!

$$\pi_i(s) = \operatorname{argmax}_a \sum_{s'} P(s'|s, a) [R(s, a, s') + \gamma U_i(s')]$$

- used after your value iteration returns with its final utility vector $\vec{U}_i$, and will generate a policy $\vec{\pi}_i$ which should approximate the optimal $\vec{\pi}^*$

4. The expected return of enforcing a fixed policy $\vec{\pi}$

$$U(s) = \sum_{s'} P(s'|s, \pi(s)) [R(s, \pi(s), s') + \gamma U(s')]$$

- a linear equation since 'max' is removed
- use linear algebra tools to solve for $\vec{U}$
- used by the policy evaluation step in policy-iteration method